GEL-Ped: An instability-triggered safeguard for short-horizon pedestrian prediction under changing crowd flows

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Abstract

Safe navigation around pedestrians depends on forecasts that arrive soon enough to change the next action of a robot, vehicle, or crowd-management system. Learned interaction models can be accurate in familiar conditions yet issue an unstable correction when crowd flows cross, merge, or enter through an unfamiliar layout. Existing work offers strong predictors, uncertainty scores, and adaptation methods, but less often specifies what a deployed predictor should output at the moment a warning appears. This paper presents GEL-Ped (Guarded Ensemble Learning for Pedestrians), a safeguard for short-horizon prediction. Independently trained graph models provide the default forecast. GEL-Ped tracks their local disagreement, compares it with an envelope learned only from complete training runs, and leaves the graph forecast unchanged while disagreement remains inside that envelope. An exceedance moves the forecast smoothly toward a feature-based HGB fallback with an explicit speed cap. The decision uses only information available at prediction time and exposes an intervention signal to the surrounding system. The evaluation distinguishes ordinary accuracy from the failure mode that motivates the safeguard through complete-run and leave-one-scene-out tests, matched routing controls, tail-error analyses, and checks of temporal causality and retraining sensitivity. GEL-Ped is therefore positioned as an auditable response to detectable model instability, not as a replacement for a modern trajectory model or a formal safety guarantee.

Keywords: pedestrian prediction, crowd dynamics, graph neural network, deep ensemble, distribution shift, uncertainty-aware transport systems

1. Introduction

A pedestrian forecast has operational value only if it arrives early enough to change a decision. A mobile robot may slow before entering a person's path, an automated vehicle may delay a turn, and a station-monitoring system may flag a developing conflict before it spreads through a crowd. In each case the immediate question is modest but time-critical: given motion observed up to the present, what movement is plausible over the next short interval? This repeated response task differs from predicting a final destination or generating several long-range routes. It places greater weight on current velocity, nearby interactions, rapid updates, and graceful behaviour when the data no longer resemble training conditions (Rudenko et al., 2020; Kothari et al., 2022).

Crowds make even this short horizon difficult. People preserve personal space, anticipate encounters, and adjust speed and passing side as a local flow changes (Moussaïd et al., 2011; Karamouzas et al., 2014). Learned interaction models can represent these coupled responses, but a good average test score is not an operating contract. A model can be accurate on most observations and still issue a rare, unusually large correction when it meets a crossing or merging pattern that was poorly represented during training. Standard uncertainty or distance scores may warn that a case is unusual, yet a score alone does not tell a motion planner which forecast to use. A permanently mixed hybrid does produce a forecast, but it also constrains the learned model when that model is working normally. Full online adaptation can recover performance after a shift, although it changes the model and requires suitable delayed feedback (Ivanovic et al., 2023; Park et al., 2024).

This leaves a practical gap between detecting a problem and acting on it. Four questions must be answered together: what observable signal can reveal an unreliable learned correction before its future error is known;

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what forecast should be used when that signal rises; which failures remain invisible to the signal; and how can the warning support the next transport or robotic decision? Uncertainty estimation, selective prediction, online adaptation, and runtime assurance each address part of this chain (Ovadia et al., 2019; Malinin et al., 2021; Geifman and El-Yaniv, 2017; Sha, 2001). They have rarely been joined and tested as one causal mechanism for repeated short-horizon pedestrian prediction.

GEL-Ped joins these pieces with a deliberately small rule. Independently trained graph predictors estimate corrections around recent velocity, and their mean is the default forecast. The method tracks how strongly they disagree for the same pedestrian using only past and present observations. This disagreement is compared with an envelope estimated from held-out complete training runs. Inside the envelope the graph retains full control. Only an exceedance reduces its weight, moving the output smoothly toward a feature-based HGB fallback with an explicit speed cap. Its inputs summarize recent motion, personal space, forward attention, and closing motion. GEL-Ped does not attempt to recognise a named facility or declare a whole scene out of distribution. It asks a narrower question that can be audited during use: are plausible versions of the deployed predictor disagreeing more than they did in supported operation?

The same design defines the limit of the method. Disagreement can reveal a *variance-type failure*, in which independently trained members react differently to the same situation. It cannot reveal a *shared-bias failure*, in which all members agree on the same wrong correction. Richer context, broader training data, a more diverse model family, or online adaptation may be required for the latter. The intervention level is consequently a warning and routing signal, not a probability of error or a certificate of safety.

The paper makes four connected contributions. First, it defines a local, past-only instability score and estimates its operating envelope without splitting frames from the same physical run across fitting and calibration. Second, it converts an exceedance into a one-sided, continuous move toward a feature-based fallback; the fallback has no mandatory weight in supported conditions. Third, it separates average accuracy from detection quality, tail risk, intervention frequency, matched routing controls, temporal-window integrity, and retraining sensitivity. Fourth, it returns both a usable velocity forecast and an intervention level that a planner can log or act on. The claim is deliberately scoped: GEL-Ped safeguards a detectable neural failure mode; it is not a universal long-horizon trajectory generator.

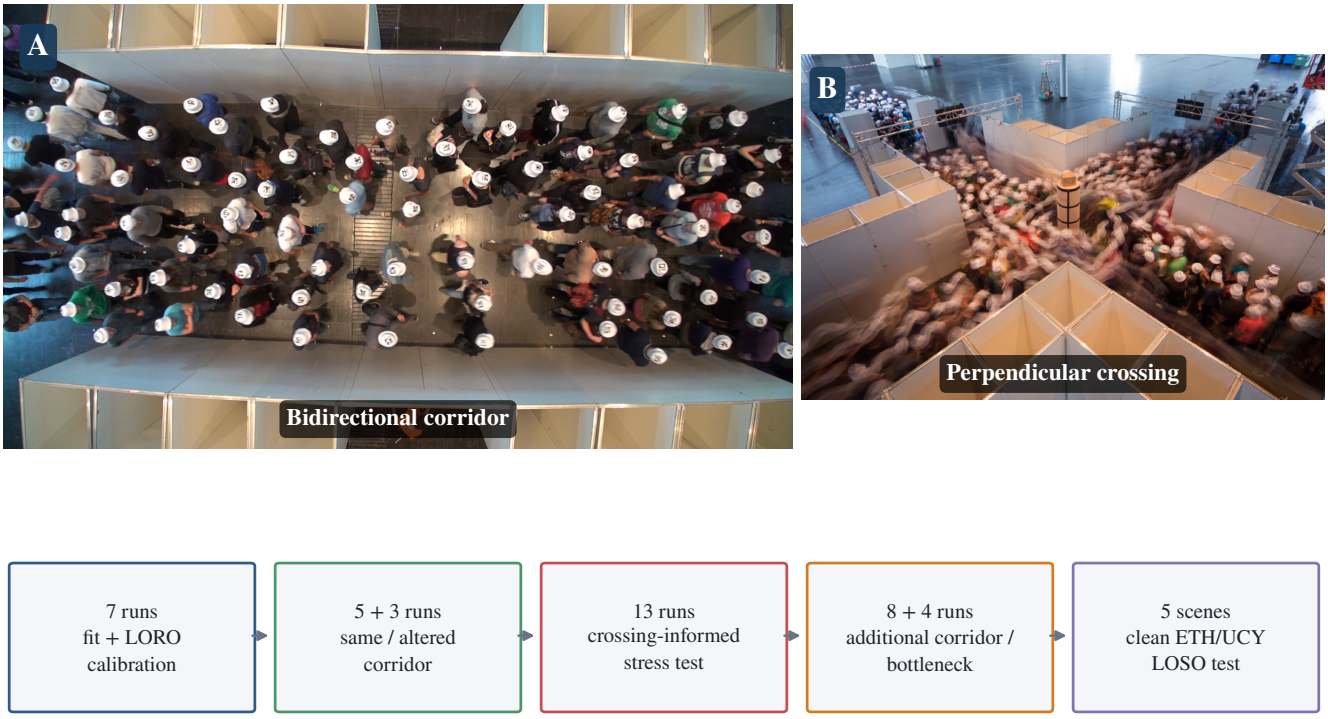


Figure 1: Empirical settings and data roles. The controlled photographs show bidirectional corridor and perpendicular crossing flows. Complete physical runs, rather than shuffled frames, define fitting, calibration, and evaluation units. Naturalistic ETH/UCY scenes and an untouched T-junction flow provide complementary transfer settings. Photographs: Juelich Pedestrian Dynamics Data Archive, CC BY 4.0; DOIs 10.34735/ped.2013.5 and 10.34735/ped.2013.4.

2. Related work and research gap

2.1. Pedestrian interaction and task scope

Pedestrian forecasting spans different tasks, from the next movement response to long-horizon multimodal paths. Social-force, collision-prediction, and velocity-obstacle models express persistence and encounter geometry in interpretable variables (Helbing and Molnár, 1995; Zanlungo et al., 2011; Karamouzas et al., 2014; van den Berg et al., 2008). Learned predictors use recurrent pooling, graphs, attention, and multimodal decoders to represent richer interactions (Alahi et al., 2016; Gupta et al., 2018; Mohamed et al., 2020; Salzmann et al., 2020). Surveys caution that results do not transfer automatically across histories, horizons, sampling rates, and output types (Rudenko et al., 2020; Kothari et al., 2022). Empirical crowd research makes the related distinction between control within an experiment and validity for the intended crowd and facility (Haghani and Sarvi, 2018; Haghani, 2023). Published errors for models that observe and predict several seconds are therefore not matched baselines for the repeated 0.8-s-history/0.4-s-response task studied here.

Recent transport research strengthens the case for structure that reflects the operating setting. Learned latent modes can make pedestrian interactions more interpretable (Liu et al., 2024); contextual and visual cues can connect crossing intention with trajectory prediction (Munir and Kucner, 2025); and a utility layer can select among learned cyclist trajectories in rare interactions (Li et al., 2023). At the aggregate level, an interpretable pedestrian-flow model can be recalibrated online as a facility changes (Mai et al., 2025). These studies improve behavioural representation or adaptation. GEL-Ped addresses a different question: what immediate point forecast should a fixed short-horizon predictor return when its own members become unstable?

2.2. Detecting unreliable predictions

Uncertainty estimates can deteriorate as test data move away from training conditions (Ovadia et al., 2019). The Shifts benchmark extends this concern to real distribution changes and promotes risk–retention evaluation, including for motion prediction (Malinin et al., 2021). Different monitors answer different questions. Mahalanobis distance measures how unusual an input or representation is relative to training data (Lee et al., 2018); ensemble disagreement asks whether plausible trained models give different answers. Ensemble-based epistemic uncertainty has supported novelty detection and recovery in autonomous driving (Filos et al., 2020). Neither score is automatically the probability that a pedestrian forecast is wrong. Its usefulness must be tested for the target shift, horizon, and error.

2.3. From a warning to an action

Selective prediction withholds cases above an uncertainty threshold and reports the trade-off between retained coverage and risk (Geifman and El-Yaniv, 2017; Cortes et al., 2016). A pedestrian planner, however, may still need a trajectory at every update. Another option is adaptation: trajectory models can update in a new environment, and adaptive conformal methods can update uncertainty sets as delayed errors arrive (Ivanovic et al., 2023; Park et al., 2024; Gibbs and Candès, 2021; Dixit et al., 2023). These methods complement rather than duplicate GEL-Ped. They learn after feedback or construct uncertainty sets; GEL-Ped provides an immediate point forecast while keeping the primary predictor fixed.

Runtime-assurance architectures supervise a capable component with a simpler response when operating conditions become unacceptable (Sha, 2001). Prediction with expert advice similarly studies how competing predictors should be weighted as their observed losses change (Herbster and Warmuth, 1998; Cesa-Bianchi and Lugosi, 2006). These foundations mean that switching or weighting is not itself new. The unresolved pedestrian-specific question is whether a causally available, local instability signal identifies the occasions on which a compact fallback is useful, beyond fixed weighting, generic input distance, or delayed expert updates.

2.4. Research gap

The literature provides strong pedestrian models, useful warning scores, principled rejection rules, online adaptation, and fallback architectures. What remains insufficiently tested is the causal chain between them for short-horizon crowd response: the monitor must detect the target failure before its label arrives; the action must still produce a usable trajectory; matched controls must distinguish a useful gate from a generally good fallback; and the blind spots of the monitor must be stated. GEL-Ped is organised around this chain. Its detector is member disagreement, its action is one-sided shrinkage toward a feature-based expert, its scope is variance-type instability rather than every form of distribution shift, and its intended use is a fast prediction loop that can consume both the forecast and the intervention level.

Table 1: Positioning by the deployment questions addressed. Detection and action are separated because an uncertainty score alone does not produce a pedestrian forecast.

Approach	Detection signal	Action after detection	Scope and use
Shift and uncertainty monitoring (Ovadia et al., 2019; Lee et al., 2018; Malinin et al., 2021; Filos et al., 2020)	confidence, feature distance, or disagreement	score, retain, recover, or request feedback	recognises conditions in which ordinary confidence may be unreliable
Selective prediction (Geifman and El-Yaniv, 2017; Cortes et al., 2016)	confidence threshold	reject or abstain	trades coverage for lower retained risk
Adaptation and online experts (Gibbs and Candès, 2021; Dixit et al., 2023; Ivanovic et al., 2023; Park et al., 2024; Herbster and Warmuth, 1998)	delayed errors or test-time information	update sets, parameters, or expert weights	recovers after feedback under a changing stream
GEL-Ped (this study)	past-only local disagreement relative to a complete-run envelope	continuously temper the graph forecast with a feature-based fallback	supplies a forecast and intervention level for repeated short-horizon response

3. GEL-Ped method

3.1. Overview and prediction task

GEL-Ped has three parts. A graph ensemble supplies the default short-horizon forecast. A compact model built from the same recent-motion and interaction cues supplies an alternative forecast. A local guard decides how much of the alternative is needed. The guard is one-sided: it cannot improve an ordinary case by averaging two models because its graph weight is exactly one until disagreement exceeds the calibration envelope.

For pedestrian i at forecast origin t, the input contains positions no later than t. The target is mean velocity over the next $\Delta = 0.4$ s. Recent velocity over $H_o = 0.8$ s provides a strong and interpretable anchor,

$$\mathbf{b}_i(t) = \frac{\mathbf{x}_i(t) - \mathbf{x}_i(t - H_o)}{H_o}, \qquad \mathbf{v}_i^+(t) = \frac{\mathbf{x}_i(t + \Delta) - \mathbf{x}_i(t)}{\Delta}. \tag{1}$$

A past-only route unit vector $\mathbf{e}_i$ defines the forward axis, and $\boldsymbol{\ell}_i = (-e_{iy}, e_{ix})$ defines the lateral axis. The primary analysis uses the direction of prior velocity; an entrance-derived axis is retained only as a controlled sensitivity. Wall fields are excluded from every primary model so that a layout-specific boundary code cannot define transfer.

3.2. Causal interaction representation

The representation uses motion cues that have a simple behavioural interpretation, without claiming that they form a complete theory of human movement. At most eight neighbours within 3 m are retained. For neighbour j, let $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, $d_{ij} = \|\mathbf{r}_{ij}\|$, and $\mathbf{u}_{ij} = \mathbf{b}_j - \mathbf{b}_i$. With body radius $r_b = 0.25$ m and interaction range $\lambda = 0.8$ m,

$$m_{ij} = \exp\left[-\frac{d_{ij} - 2r_b}{\lambda}\right]. \tag{2}$$

Three vector fields summarize the current neighbourhood. The isotropic field points away from nearby people and weights neighbour j by m_{ij}. The forward field gives more weight to neighbours in front, using $[\max(\mathbf{r}_{ij}^{\mathsf{T}}\mathbf{e}_i/d_{ij}, 0)]^2$. The closing field additionally gives more weight to neighbours who are approaching, using $\min\{[-\mathbf{u}_{ij}^{\mathsf{T}}\mathbf{r}_{ij}/d_{ij}]_+, 3\}$. Distances are numerically lower-bounded by 10^{-8} m.

Recent velocity, route direction, and the three fields are projected onto $(\mathbf{e}_i, \boldsymbol{\ell}_i)$, giving ten focal scalars. The graph also receives a seven-value record for each retained neighbour: two local position components, two local relative-velocity components, distance, closing speed, and a presence indicator. Missing slots are masked. This separation lets the graph reason about individual neighbours while the fallback sees only compact summaries.

Table 2: Main notation. All quantities used by the deployed predictor are computed from times no later than the forecast origin t.

Symbol	Meaning	Fixed role in this study
$\mathbf{x}_i(t)$	planar position of pedestrian i	observed through t
$\mathbf{b}_i(t)$	recent-velocity anchor	displacement over $H_o = 0.8$ s
$\mathbf{v}_i^+(t)$	future mean velocity target	mean over $\Delta = 0.4$ s; never used by the deployed guard
$\mathbf{e}_i, \boldsymbol{\ell}_i$	past-only forward and lateral axes	local coordinates for features and residuals
$\widehat{\mathbf{v}}_i^{G,m}$	prediction of graph member m	$M = 5$ members in the primary ensemble
$\widehat{\mathbf{v}}_i^F$	feature-based fallback prediction	same anchor and focal inputs as the graph
$s_i(t), S_i(t)$	instantaneous and time-smoothed disagreement	local to one pedestrian and reset between scenes
τ, q_{IQR}	calibration threshold and scale	finite-sample 90th percentile and interquartile range
$g_i(t)$	graph weight	one inside the envelope; decreases only after an exceedance

3.3. Graph ensemble and feature-based HGB fallback

Each graph member transforms the seven variables of every neighbour with a shared two-layer rectified encoder, attention-pools the resulting messages, and joins the pooled message with the ten focal scalars. The edge encoder is $7 \to 48 \to 48$; a $48 \to 1$ attention score pools valid neighbours; and the node head is $58 \to 96 \to 48 \to 2$. The two outputs are forward and lateral corrections around recent velocity,

$$\widehat{\mathbf{v}}_i^{G,m} = \mathbf{b}_i + \delta_{i,\parallel}^{G,m}\mathbf{e}_i + \delta_{i,\perp}^{G,m}\boldsymbol{\ell}_i, \qquad m = 1, \ldots, M, \quad M = 5. \tag{3}$$

The graph mean is $\bar{\mathbf{v}}_i^G = M^{-1}\sum_m \widehat{\mathbf{v}}_i^{G,m}$. With wall inputs removed, each member has 13,203 trainable parameters. Runs contribute equally to the loss even when they contain different numbers of origins. If run k has n_k training origins, then

$$\mathcal{L} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k}\left\|\widehat{\mathbf{v}}_{ki}^G - \mathbf{v}_{ki}^+\right\|_2^2. \tag{4}$$

Each member is trained for 25 epochs with AdamW, learning rate 10^{-3}, weight decay 10^{-3}, and gradient-norm cap 5. The architecture and budget are frozen from complete-run calibration rather than chosen on a test regime.

The fallback uses the same ten focal scalars and anchor but no individual-neighbour message passing. Separate histogram-gradient-boosting regressors predict the two residual components,

$$\widehat{\mathbf{v}}_i^F = \mathbf{b}_i + \delta_{i,\parallel}^F\mathbf{e}_i + \delta_{i,\perp}^F\boldsymbol{\ell}_i. \tag{5}$$

Each regressor uses 200 boosting iterations, at most 31 leaves, minimum leaf size 50, learning rate 0.05, and L_2 regularization 1.0. Both experts apply the same explicit 3 m/s output-speed cap. The fallback is chosen for finite piecewise-constant extrapolation and compact inputs, not because a tree is a complete behavioural model. Persistence, personal space, forward attention, and closing motion enter through its features; destination choice, group membership, and visual context do not.

3.4. Local instability guard

The guard begins with the root-mean-square disagreement of graph members for one pedestrian,

$$s_i(t) = \left[\frac{1}{M}\sum_{m=1}^{M}\left\|\widehat{\mathbf{v}}_i^{G,m}(t) - \bar{\mathbf{v}}_i^G(t)\right\|_2^2\right]^{1/2}. \tag{6}$$

The score is never averaged across different pedestrians. Forecast origins are 0.4 s apart. The smoothed score $S_i(t)$ averages the current value and up to four preceding values for the same pedestrian whose origins fall in the trailing 1.6 s. State is reset when the pedestrian or scene changes. Smoothing is performed on the

chronological sequence before any training subsampling, so the window represents elapsed time rather than an arbitrary number of retained rows.

Calibration is leave-one-complete-run-out. For each fold, graph members and the fallback are fitted on six runs, then $S_i(t)$ is computed chronologically for every eligible origin in the seventh. Pooling only these out-of-fold scores gives N calibration values $S_{(1)} \le \cdots \le S_{(N)}$. The envelope threshold uses the finite-sample 90th-percentile convention

$$\tau = S_{(k)}, \qquad k = \min\{N, \lceil (N+1)0.90 \rceil\}, \tag{7}$$

and the scale is $q_{\mathrm{IQR}} = \max\{Q_{0.75} - Q_{0.25}, 10^{-6}\}$ m/s. The deployed graph weight is

$$g_i(t) = \exp\left[-\frac{(S_i(t) - \tau)_+}{q_{\mathrm{IQR}}}\right], \qquad \widehat{\mathbf{v}}_i(t) = g_i(t)\bar{\mathbf{v}}_i^G(t) + (1 - g_i(t))\widehat{\mathbf{v}}_i^F(t). \tag{8}$$

Thus $g_i(t) = 1$ throughout the calibration envelope. The fallback has no mandatory weight, and the guard cannot improve an ordinary prediction merely by fixed ensembling. Linear and logistic weight maps are sensitivity checks; the exponential map above is the frozen primary rule. Figure 2 summarizes the calculation.

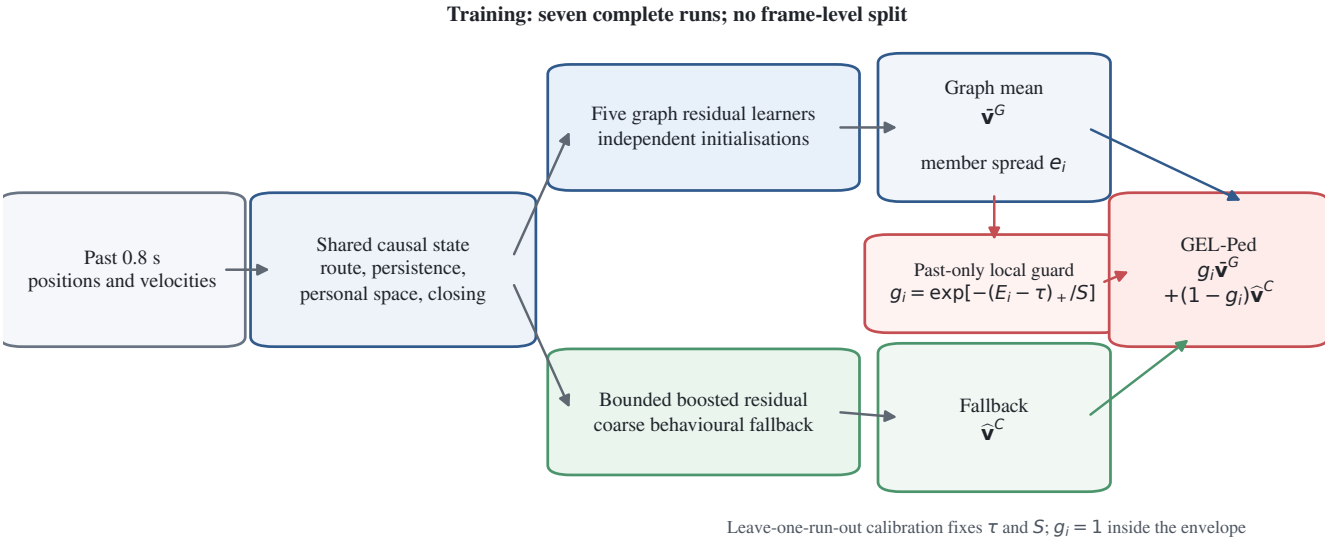


Figure 2: GEL-Ped. Five anchored graph learners provide the default mean and a local disagreement score. Leave-one-complete-run-out calibration fixes the score envelope. Only an exceedance moves the forecast toward the feature-based HGB fallback with an explicit speed cap. All deployed signals use past or present observations.

4. Experimental design

4.1. Data roles and test firewall

The controlled archive contains 25-Hz bidirectional-corridor and right-angle-crossing trajectories (Boltes and Seyfried, 2013; Boltes et al., 2020). Seven complete corridor runs spanning entrance widths and occupancies form the fitting and cross-fitting set. Five familiar tests retain the 22-m corridor and 5-m exit, while three altered tests change corridor length and/or exit arrangement. Crossing observations informed mechanism development and are treated as a development-informed stress test, not as untouched confirmation. The immutable run record in Appendix Appendix A documents the physical units of analysis.

Eight unidirectional-corridor and four bottleneck runs from the Tordeux/Juelich archive are descriptive transfer settings (DOI 10.5281/zenodo.1054017). Their 16-Hz tracks are placed on a 10-Hz grid using only the two most recent source observations available at each grid time. This past-only linear update never reads a future source point and does not bridge an internal tracking gap; it keeps the 0.8-s history, 0.4-s target, and 0.4-s origin spacing exact. A separately archived T-junction flow (DOI 10.26165/JUELICH-DATA/YNCHYF) is reserved as an untouched, one-shot shift test: no architecture, threshold, cap, or feature definition is updated after its outcome is opened.

Naturalistic transfer uses the standard five-scene ETH/UCY leave-one-scene-out protocol distributed with Trajectron++ (Pellegrini et al., 2009; Salzmann et al., 2020). For each held scene, the remaining training

scenes fit the experts and the supplied validation split calibrates the guard; the held scene is evaluated once. At 2.5 Hz, the 0.8-s history comprises two intervals and the 0.4-s target one interval. All data roles are assigned at the level of complete physical runs or scenes, never shuffled frames.

Eligibility is causal. Origins require the full 0.8-s observed history and 0.4-s target for offline scoring. A 3 m/s quality limit is applied only to speed measured in the observed history; no target-speed or other future-label filter is used. Model fitting samples at most 6,000 eligible origins per run with a fixed seed. Evaluation uses every eligible origin in chronological order. The time-based guard is constructed on that full chronological sequence before any fitting sample is drawn. The deployed predictor never receives a future position, regime label, or test error.

4.2. Comparators and selection discipline

The primary comparators are constant velocity, one graph member, the five-member graph mean, the coordinate-wise median of those members, and the feature-based HGB fallback. Gate controls use the same origin set, graph predictions, fallback predictions, and selected member draw. They include fixed expert weights; a permutation control that preserves the weight distribution but breaks its link to the case; Mahalanobis distance on the same focal state; graph–fallback disagreement; a learned gate trained only on calibration folds; and a speed-only gate. A future-error oracle is labelled explicitly as an unattainable upper bound. A delayed online expert-weighting comparator updates only after the corresponding target becomes observable, so it never uses present-origin error prematurely.

Architecture and training choices are frozen on the seven calibration runs. The primary ensemble contains five members. Retraining sensitivity draws 30 ensembles from a pre-trained pool of 25 independently seeded members and separately checks ensemble sizes 3, 5, 10, and 20. A compact transformer replaces the graph encoder in a backbone-robustness check while retaining the same causal inputs, target, splits, fallback, and guard definition. Published 3.2-s/4.8-s ETH/UCY errors are not mixed with the present 0.8-s/0.4-s task; external methods are compared only when the observation, horizon, inputs, and output are matched.

4.3. Endpoints and inference

The primary accuracy endpoint is vector RMSE in m/s, computed within each complete run and then averaged with equal run weight. Displacement error and angular and speed components are secondary. Tail behaviour is reported by the 90th, 95th, and 99th error percentiles, CVaR at 95% and 99%, and the rate of errors above 0.5 m/s. Detection is evaluated against the largest 5% of graph errors using AUROC and AUPRC, together with risk–retention curves and area under the risk–retention curve. These quantities distinguish a useful warning from an improvement caused only by the fallback.

All paired inference operates on complete-run differences. Exact sign-flip tests and multiplicity correction are used where the predeclared family has sufficient runs; run bootstrap intervals resample runs, not individual origins. Practical equivalence to the graph is assessed with both an absolute margin of 0.002 m/s and a relative margin of 1%. The fixed, permuted, matched, and oracle controls are reported at the same intervention rate or weight distribution wherever the comparison requires it.

The guard's intervention level is defined by $(S_i(t) - \tau)/q_{\mathrm{IQR}}$; values above two denote a score more than two calibration IQRs beyond the envelope. Local density, neighbour count, and observed speed are analysed separately so that disagreement is not mistaken for a density or speed proxy. Decomposition by ensemble draw separates errors of the graph expert, fallback expert, and routing decision.

Recursive rollouts feed each model its own predicted state at 0.4-s increments while replaying observed neighbour fields. The proximity screen likewise holds other trajectories fixed. Both are diagnostic consequence audits rather than a reciprocal simulator or a closed-loop collision-avoidance claim.

5. Results

5.1. Where the guard helps—and where it does not

Table 3 gives the three original regimes. The graph ensemble already improves on a single initialization. GEL-Ped is effectively unchanged in familiar and altered corridors. In crossing, it changes RMSE from 0.3543 to 0.3463 m/s, a 2.3% mean reduction. The coarse HGB is marginally lower at 0.3455 m/s; the 0.0007 m/s difference from GEL-Ped is not significant. The contribution is therefore not that the guard beats every component in every regime. It preserves graph accuracy where the coarse model is 5–7% worse, while approaching the coarse model when graph members become unstable.

Table 3: Complete-run vector RMSE (m/s): mean [minimum, maximum] across physical runs. Bold marks the lowest mean in a column; ranges are descriptive, not confidence intervals.

Model	Familiar (5)	Altered (3)	Crossing (13)
Constant velocity	0.2346 [0.2020, 0.2469]	0.2302 [0.2255, 0.2328]	0.3745 [0.3514, 0.3996]
Single graph network	0.1624 [0.1439, 0.1750]	0.1560 [0.1513, 0.1600]	0.3580 [0.3225, 0.3911]
Five-member graph ensemble	0.1596 [0.1400, 0.1727]	**0.1530 [0.1479, 0.1572]**	0.3543 [0.3183, 0.3899]
Coordinate-median ensemble	0.1600 [0.1407, 0.1730]	0.1536 [0.1486, 0.1580]	0.3520 [0.3186, 0.3854]
Anchored coarse HGB	0.1710 [0.1477, 0.1904]	0.1611 [0.1557, 0.1651]	**0.3455 [0.3213, 0.3849]**
GEL-Ped	**0.1595 [0.1400, 0.1720]**	0.1531 [0.1480, 0.1574]	0.3463 [0.3180, 0.3851]

The additional configurations in Table 4 reinforce that trade-off. Relative to the graph ensemble, GEL-Ped changes mean RMSE by +0.0002 m/s in bottlenecks and −0.0003 m/s in unidirectional flow. Relative to coarse HGB, it is better by 0.0159 and 0.0206 m/s. Thus replacing the graph globally with a bounded tree would solve one crossing average at the cost of materially worse predictions in four other controlled regimes.

Table 4: Additional Juelich configurations not used for fitting: complete-run mean RMSE (m/s). The native 16-Hz data use 0.8125 s of history and a 0.375-s target.

Model	Bottleneck (4)	Unidirectional (8)
Constant velocity	0.2636	0.2594
Graph ensemble	**0.2470**	0.2396
Coordinate-median ensemble	0.2480	0.2402
Coarse HGB	0.2631	0.2599
GEL-Ped	0.2472	**0.2393**

5.2. The crossing mean, its outlier, and the mechanism

All 13 crossing differences favour GEL-Ped, and the exact test against the graph remains significant after the complete 15-test Holm correction ($p_{\text{adj}} = 0.0037$). The distribution matters more than the headline: the mean difference is −0.0080 m/s, the median is −0.0022 m/s, and leave-one-run-out means range from −0.0087 to −0.0045 m/s. Removing run `d_1` leaves a 1.3% mean reduction. Full per-run values are in Table Appendix B.1; all fifteen tests appear in Table Appendix B.2.

Run `d_1` diagnoses the intended failure mode. Four graph members have RMSE between 0.3519 and 0.4176 m/s, but the third reaches 0.8396 m/s. Their prediction mean gives 0.3899 m/s and the robust coordinate median 0.3593 m/s. GEL-Ped detects the local spread and reaches 0.3396 m/s, close to the coarse fallback at 0.3366 m/s. Across crossing, the highest instability stratum (> 2 calibration IQR, 2,875 forecasts) has graph RMSE 0.5654 m/s and GEL-Ped RMSE 0.4665 m/s. The benefit is therefore concentrated in a small neural tail rather than distributed evenly across all pedestrians.

Figure 3 makes both facts visible: the run-level mean is outlier-sensitive, while the instability strata show a mechanism that cannot be explained by simple coordinate-median aggregation alone.

5.3. Naturalistic leave-one-scene-out tests

Table 5 reports all five standard scenes. GEL-Ped is slightly worse than the graph on ETH, ZARA1, and ZARA2, nearly equal on HOTEL, and better on UNIV. The scene-macro RMSE is 0.2262 versus 0.2267 m/s, a 0.2% descriptive reduction; GEL-Ped wins two of five scenes and the exact paired p value is 0.9375. This is the cleanest external result and does not support a broad accuracy claim.

The realized mean graph weights are 0.699 (ETH), 0.994 (HOTEL), 0.898 (UNIV), 0.999 (ZARA1), and 0.986 (ZARA2). For each held scene, τ and S are recalibrated on the supplied validation split of the other scenes; no Juelich threshold is transferred. The result shows that the guard can remain mostly inactive under naturalistic leave-one-scene-out transfer, but the lower ETH weight does not reliably imply that the fallback is superior.

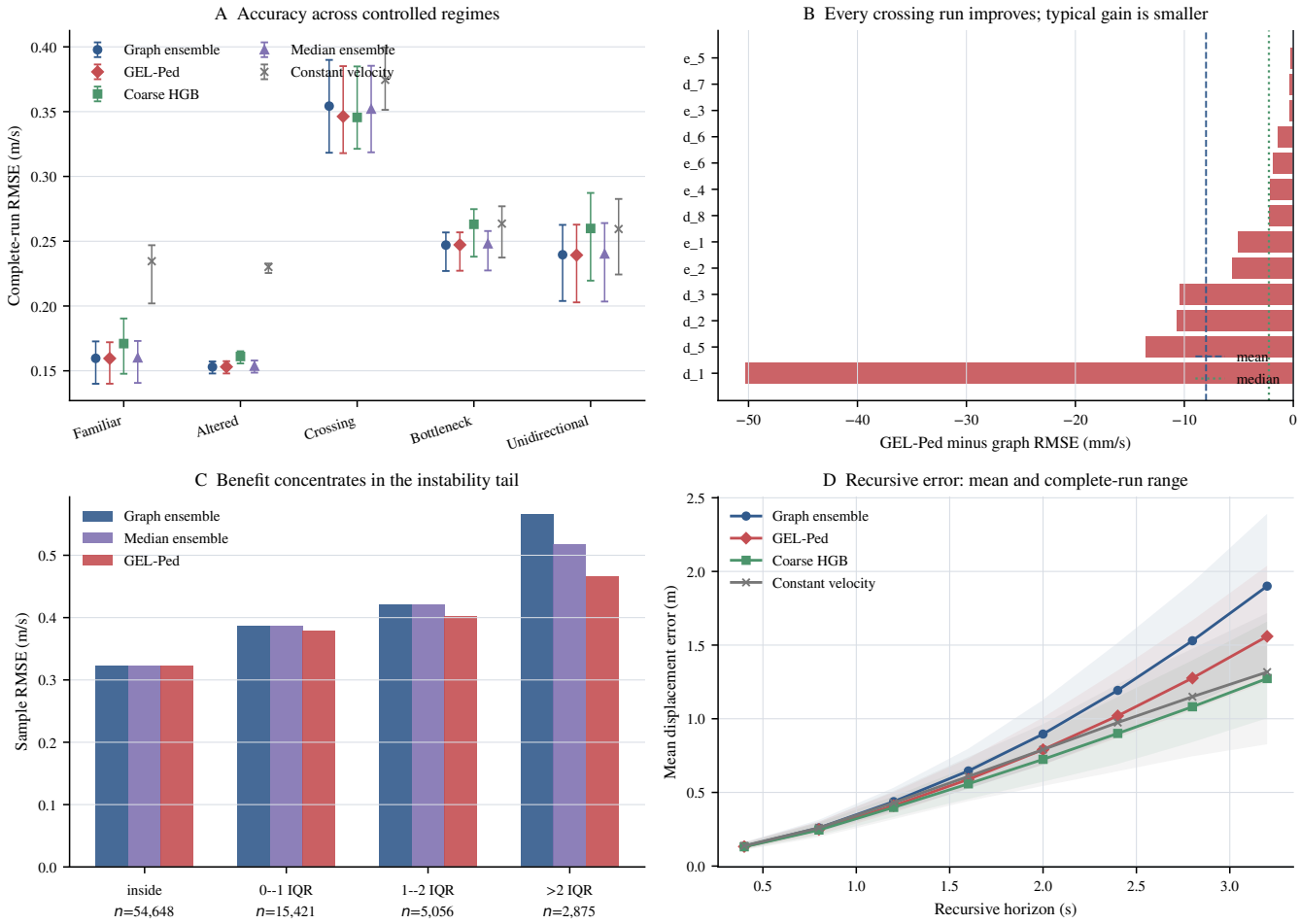


Figure 3: Final evidence. (A) Run means and complete-run ranges; CV denotes constant velocity. (B) Crossing paired differences, with mean and median shown separately. (C) Sample RMSE by calibration-normalized instability; n is the number of forecasts in each bin. (D) Recursive displacement error; lines are complete-run means and shaded regions are complete-run ranges, not confidence intervals.

Table 5: ETH/UCY leave-one-scene-out vector RMSE (m/s), reconstructed as the same 0.8-s-history/0.4-s-target task. Macro averages weight the five scenes equally.

Model	ETH	HOTEL	UNIV	ZARA1	ZARA2	Macro
Graph ensemble	**0.4111**	0.2433	0.1807	**0.1389**	**0.1594**	0.2267
Coarse HGB	0.4173	0.2461	**0.1743**	0.1413	0.1614	0.2281
GEL-Ped	0.4137	**0.2432**	0.1756	0.1390	0.1596	**0.2262**

5.4. Operational and sensitivity audits

The crossing collision screen contains 73,051 true conflict events and 4,949 non-conflict events. GEL-Ped records 72,265 true positives, 547 false positives, 786 misses, and 4,402 true negatives. The graph ensemble records 72,243, 556, 808, and 4,393. Miss rate therefore changes from 1.106% to 1.076% (2.7% relative), while false-alarm rate changes from 11.23% to 11.05%. Run-level miss-rate differences range from −0.212 to +0.018 percentage points. The previously similar rounded false-alarm values were coincidental, not threshold matching.

Frozen crossing sensitivity is not knife-edge. Across threshold quantiles 0.80, 0.90, and 0.95 and scale multipliers 0.5, 1, and 2, mean RMSE lies between 0.3452 and 0.3478 m/s. A five-observation, future-label oracle reaches 0.3413 m/s, leaving only 0.0050 m/s mean headroom. Ensemble spread correlates with speed in crossing (mean within-run Spearman $\rho = 0.433$), but much less in other regimes; occupancy and neighbour-count correlations do not reproduce the error-tail pattern. These are clustered descriptive correlations, not 78,000 independent tests.

The all-speed audit removes the future-label exclusion. Crossing RMSE becomes 0.3615 for GEL-Ped and 0.3667 m/s for the graph; GEL-Ped wins 11 of 13 runs, with mean and median differences of −0.0052 and −0.0019 m/s. With the ETH-style prior-velocity route axis on Juelich, absolute errors rise, but the crossing

comparison remains 0.3727 versus 0.3868 m/s. The guard mechanism is therefore not an artefact of route labels, although the entrance route improves absolute accuracy.

Recursive rollouts amplify the difference: complete-run mean displacement errors for graph/GEL-Ped are 0.259/0.249 m at 0.8 s, 0.646/0.589 m at 1.6 s, and 1.900/1.559 m at 3.2 s. This is a 4.1%, 8.8%, and 17.9% reduction relative to the graph. However, coarse HGB and constant velocity reach 1.273 and 1.317 m at 3.2 s, both below GEL-Ped. Recursively applying a one-step response model is therefore not a substitute for a dedicated long-horizon predictor; the audit only shows that the guard limits compounding neural error.

6. Discussion

6.1. Why not use the neural ensemble alone?

In supported flow, there is almost no reason not to. Mean g is 0.977 in familiar corridors, 0.960 under altered geometry, 0.995 in bottlenecks, and 0.955 in unidirectional flow. GEL-Ped is consequently within 0.0003 m/s of the graph ensemble in these regimes. The practical reason for the guard is narrower: a graph member can produce a rare, very large correction under a changed interaction pattern, and simple averaging or coordinate medians do not always remove it. GEL-Ped converts the same-family spread into a bounded correction without imposing a permanent fallback weight.

The comparison with coarse HGB is equally important. Coarse HGB slightly wins the crossing mean, so the paper does not present geometry as the cause of that gain. But choosing HGB globally increases error by 5–8% in the other controlled regimes. GEL-Ped supplies an operating point close to the better expert without using future error. A graph–coarse disagreement guard is even lower in crossing (0.3443 m/s) but degrades on the additional configurations and was not selected after viewing test results. The ensemble-spread rule has the clearest pre-deployment interpretation and the smallest in-support intervention.

6.2. Transportation use and limits

For a mobile robot or automated vehicle, g_i can be exposed alongside the velocity forecast. A downstream system may use persistent low values to apply a conservative planning policy, request sensor or model review, or trigger site recalibration. For stations and terminals, spatially clustered interventions can identify where the calibrated predictor is no longer behaving consistently. These are possible uses of the signal, not effects implemented or certified by the predictor itself.

Several limits remain. Crossing influenced development, and the extra Juelich configurations are descriptive. The five ETH/UCY scenes provide cleaner external evidence, but they show no significant mean advantage. Member disagreement detects variance-type instability but is blind when all members share the same error. The output is unimodal, lacks explicit destinations, groups, gaze, and visual context, and the fallback is a compact feature-based regressor rather than a theory of human behaviour. The collision screen replays neighbours rather than modelling reciprocal response. A calibration-residual disk is cross-conformal-like rather than ordinary split conformal because folds overlap and samples within runs are dependent. Its nominal 90% coverage falls to 35.9–52.2% across crossing instability strata; empirical guard-conditional radii improve stratified coverage but do not restore a finite-sample guarantee under shift. Uncertainty-set coverage is therefore not a contribution of GEL-Ped: after a topology change, the intervention should be treated as a warning or a reason to recalibrate, not as a valid 90% interval or a safety certificate.

7. Conclusion

GEL-Ped turns one observable neural failure signal into a simple deployment action. A graph ensemble remains fully responsible while its members agree within a complete-run calibration envelope. When local spread rises, its prediction moves smoothly toward a feature-based fallback. This design is easier to audit than a gate trained to recognize named regimes and less restrictive than assigning a fixed weight to a hand-crafted model.

The results support a precise value proposition. The method does not materially improve ordinary in-support averages, and a coarse boosted model is marginally best in perpendicular crossing. GEL-Ped is useful because that coarse model is substantially worse elsewhere, while the guarded ensemble preserves graph accuracy and suppresses its small but consequential instability tail. The mean crossing gain is significant but outlier-sensitive; the clean ETH/UCY macro gain is negligible and non-significant. For publication and deployment, the method should therefore be understood as a transparent neural safeguard for short-horizon crowd response, not as a universal trajectory-prediction replacement.

Data availability

The bidirectional-corridor trajectories are available at DOI 10.34735/ped.2013.5; the perpendicular-crossing trajectories at DOI 10.34735/ped.2013.4; and the additional Juelich/Tordeux configurations at DOI 10.5281/zenodo.1054017. ETH/UCY files follow the Trajectron++ repository at commit `1031c7bd1a444273af378c1ec1dcca907ba59830`. Raw files are not redistributed.

Code availability

Source code, immutable run partitions, calibration records, complete-run outputs, tests, and figure scripts will be released under a versioned public tag and archived with a persistent identifier before publication. The release will record the software version and random seeds needed to reproduce each table and figure.

Ethics statement

This study reuses publicly available, de-identified pedestrian trajectories and does not access participant identities.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Run partition and selection record

Table Appendix A.1 documents the full 7/5/3 corridor partition. Calibration runs were selected to span the available entrance widths and low-to-high occupancy. Familiar tests retain the 22-m corridor and 5-m exit. "Altered" is defined before modelling as a changed corridor length and/or exit arrangement. Nominal mean density divides mean frame occupancy by the documented 4-m corridor width and corridor length; it is a global facility statistic, not a Voronoi local density.

Table Appendix A.1: Complete-run partition and physical descriptors. A dash in exit width denotes the open terminal arrangement recorded as zero in the archive metadata.

Run	Role	Ped.	Duration (s)	Entrance (m)	Length (m)	Exit (m)	Nominal density	Forecasts
a_01	calibration	377	93.0	1.8	22	5.0	0.69	6000
a_02	calibration	521	142.3	3.0	22	5.0	1.10	6000
b_01	calibration	141	76.1	0.6	22	5.0	0.16	1924
b_03	calibration	480	129.8	1.2	22	5.0	0.42	6000
b_05	calibration	643	172.6	2.4	22	5.0	1.07	6000
b_07	calibration	606	162.9	4.0	22	5.0	1.34	6000
b_09	calibration	483	119.3	2.4	22	5.0	0.65	6000
a_03	familiar	541	140.7	4.0	22	5.0	1.22	6000
b_02	familiar	259	93.2	0.9	22	5.0	0.26	3989
b_04	familiar	743	189.8	1.8	22	5.0	0.97	6000
b_06	familiar	830	248.8	3.0	22	5.0	1.29	6000
b_10	familiar	736	207.7	4.0	22	5.0	1.02	6000
a_04	altered	544	163.0	4.0	17	2.5	1.76	6000
a_05	altered	559	169.8	4.0	12	–	2.36	6000
b_08	altered	703	230.2	4.0	12	–	2.81	6000

Table Appendix B.1: Crossing complete-run RMSE (m/s), graph-member seed SD, and realized mean graph weight.

Run	Graph mean	Median ensemble	Coarse HGB	GEL-Ped	GEL–Graph	Seed SD / $\bar{g}$
d_1	0.3899	0.3593	0.3366	0.3396	−0.0503	0.2071 / 0.812
d_2	0.3689	0.3701	0.3559	0.3582	−0.0106	0.0024 / 0.712
d_3	0.3676	0.3681	0.3489	0.3571	−0.0104	0.0028 / 0.728
d_5	0.3630	0.3601	0.3464	0.3494	−0.0136	0.0211 / 0.856
d_6	0.3338	0.3339	0.3304	0.3324	−0.0014	0.0020 / 0.927
d_7	0.3183	0.3186	0.3213	0.3180	−0.0003	0.0025 / 0.926
d_8	0.3266	0.3273	0.3266	0.3244	−0.0022	0.0010 / 0.900
e_1	0.3352	0.3355	0.3281	0.3301	−0.0051	0.0030 / 0.866
e_2	0.3732	0.3736	0.3687	0.3676	−0.0056	0.0024 / 0.814
e_3	0.3855	0.3854	0.3849	0.3851	−0.0004	0.0027 / 0.868
e_4	0.3614	0.3613	0.3614	0.3593	−0.0021	0.0024 / 0.911
e_5	0.3416	0.3416	0.3442	0.3414	−0.0002	0.0024 / 0.926
e_6	0.3408	0.3409	0.3386	0.3389	−0.0018	0.0021 / 0.884

Table Appendix B.2: All fifteen predeclared paired tests. Differences are GEL-Ped minus comparator in m/s; negative favours GEL-Ped.

Regime	Comparator	Wins/n	Mean diff.	Median diff.	Exact p	Holm p
Familiar	Graph mean	2/5	−0.00010	0.00004	0.8750	1.0000
Familiar	Median ensemble	4/5	−0.00050	−0.00061	0.1875	1.0000
Familiar	Coarse HGB	5/5	−0.01148	−0.00945	0.0625	0.6875
Altered	Graph mean	1/3	0.00008	0.00002	0.5000	1.0000
Altered	Median ensemble	3/3	−0.00055	−0.00053	0.2500	1.0000
Altered	Coarse HGB	3/3	−0.00800	−0.00774	0.2500	1.0000
Crossing	Graph mean	13/13	−0.00801	−0.00224	0.00024	0.00366
Crossing	Median ensemble	13/13	−0.00572	−0.00293	0.00024	0.00366
Crossing	Coarse HGB	5/13	0.00074	0.00027	0.4458	1.0000
Unidirectional	Graph mean	3/8	−0.00029	0.00007	0.3359	1.0000
Unidirectional	Median ensemble	8/8	−0.00093	−0.00089	0.00781	0.1016
Unidirectional	Coarse HGB	8/8	−0.02059	−0.02163	0.00781	0.1016
Bottleneck	Graph mean	0/4	0.00020	0.00016	0.1250	1.0000
Bottleneck	Median ensemble	4/4	−0.00082	−0.00092	0.1250	1.0000
Bottleneck	Coarse HGB	4/4	−0.01585	−0.01719	0.1250	1.0000

Table Appendix C.1: Guard and OOD controls: mean RMSE (m/s). ETH/UCY is the equal-scene macro average. These variants were not selected on test performance.

Model	Crossing	Bottleneck	Unidirectional	ETH/UCY
Graph ensemble	0.3543	**0.2470**	0.2396	0.2267
Graph–coarse disagreement guard	**0.3443**	0.2486	0.2419	0.2264
Graph–CV disagreement guard	0.3526	**0.2470**	0.2396	0.2273
Prospective fallback, spread guard	0.3456	0.2489	0.2418	0.2256
Learned mixture-of-experts gate	0.3469	0.2515	0.2451	0.2255
Mahalanobis OOD guard	0.3481	0.2488	0.2418	**0.2252**
GEL-Ped	0.3463	0.2472	**0.2393**	0.2262

Appendix B. Crossing runs and complete Holm family

Appendix C. Guard controls, capacity, and uncertainty

The Mahalanobis and learned gates slightly lower the external macro mean, but neither was the prespecified primary mechanism. The graph–coarse disagreement guard is the crossing winner yet intervenes more on non-crossing data. GEL-Ped is retained because its signal is internal to the neural family, its in-support weight is exactly one, and its intervention is directly attributable to ensemble instability.

The residual uncertainty audit is empirical. A global 90% radial quantile calibrated from overlapping leave-one-run-out residuals is cross-conformal-like; dependent frames and overlapping fits prevent an ordinary split-conformal guarantee. Mondrian radii estimated by guard stratum are 0.244, 0.290, 0.339, and 0.364 m/s from inside to > 2 IQR. Crossing coverages are 50.7%, 51.8%, 59.7%, and 56.9%, respectively, compared with global-radius coverage of 52.2%, 42.5%, 41.0%, and 35.9%. Conditional calibration helps the tail but does not

Table Appendix C.2: Calibration-only graph capacity/epoch sweep. RMSE averages seven leave-one-complete-run-out folds.

Width	Epochs	Parameters	Calibration RMSE (m/s)
32	25	6,019	0.16453
48	25	13,395	**0.16398**
48	45	13,395	0.16560
48	75	13,395	0.16638
96	45	51,939	0.16560
96	75	51,939	0.16810

repair topology-shift coverage.

Appendix D. Encoding and label-filter sensitivities

Table Appendix D.1: Frozen encoding audits: mean complete-run RMSE (m/s). The all-speed audit removes the future-label filter; the prior-velocity axis replaces the entrance route with recent displacement and is independently cross-fitted.

Encoding	Regime	Graph ensemble	Coarse HGB	GEL-Ped
Primary entry axis	Familiar	0.1596	0.1710	0.1595
	Altered	0.1530	0.1611	0.1531
	Crossing	0.3543	0.3455	0.3463
Prior-velocity axis	Familiar	0.1730	0.1916	0.1735
	Altered	0.1678	0.1816	0.1683
	Crossing	0.3868	0.3652	0.3727
All observed speeds	Familiar	0.1596	0.1710	0.1595
	Altered	0.1530	0.1611	0.1531
	Crossing	0.3667	0.3606	0.3615

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